



FINDING THE ZONE: RE-EXAMINING THE HOT HAND IN BASEBALL

Insights from Streak Tests, Rolling Windows, and Pitch-Location Data

- Presented by Shenwei Zhang
- Case Study: Michael Harris II

WHAT IS THE “HOT HAND”?

- Belief

recent success → higher future success

- Common in sports

“on fire,”
“locked in,”
“in the zone”

- real performance change or RANDOMNESS?





CLASSICAL VIEW: HOT HAND = ILLUSION

Gilovich, Vallone & Tversky (1985)

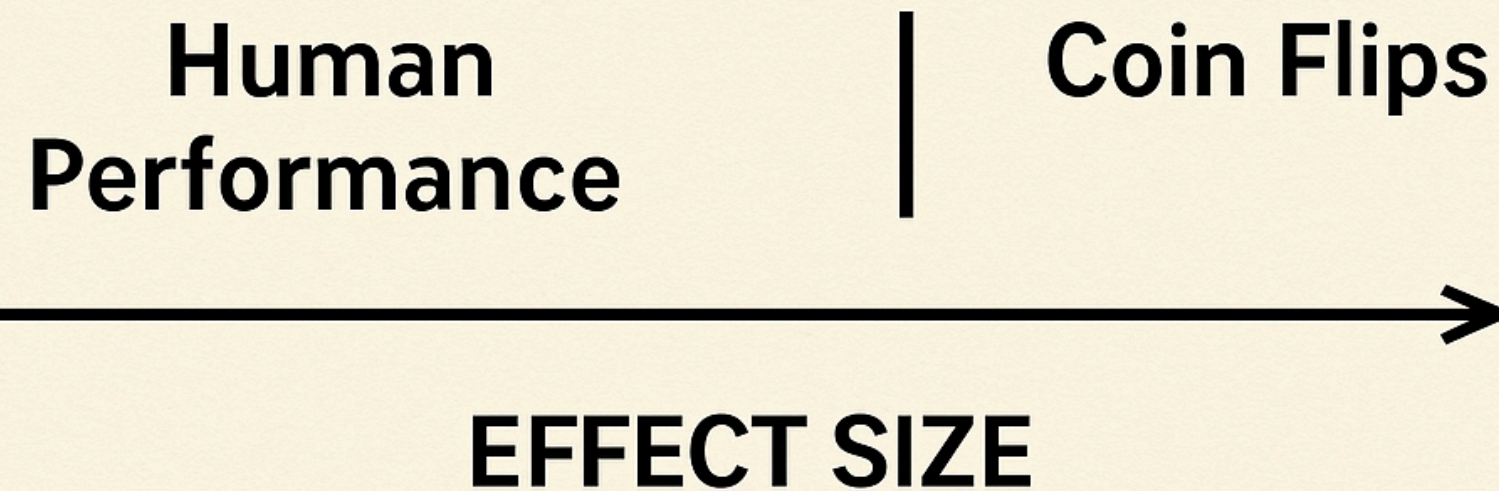
Streaks occur naturally in random sequences

Fans misread clustering → representativeness bias

Conclusion: illusion of pattern-seeking



UPDATED LITERATURE: SMALL BUT REAL EFFECTS



Later research found small positive recency effects

Not dramatic, but statistically detectable

Simple streak tests often too noisy

Move toward nuanced, player-specific patterns

OVERVIEW- DATA

Data:

MLB Statcast (2022–2025)

~4000 pitches, all plate appearances

Pitch speed, location, events, game date

Harris only

METHODS



01

Rolling Batting Average
(25 PA WINDOWS)

02

Swing-location heatmaps

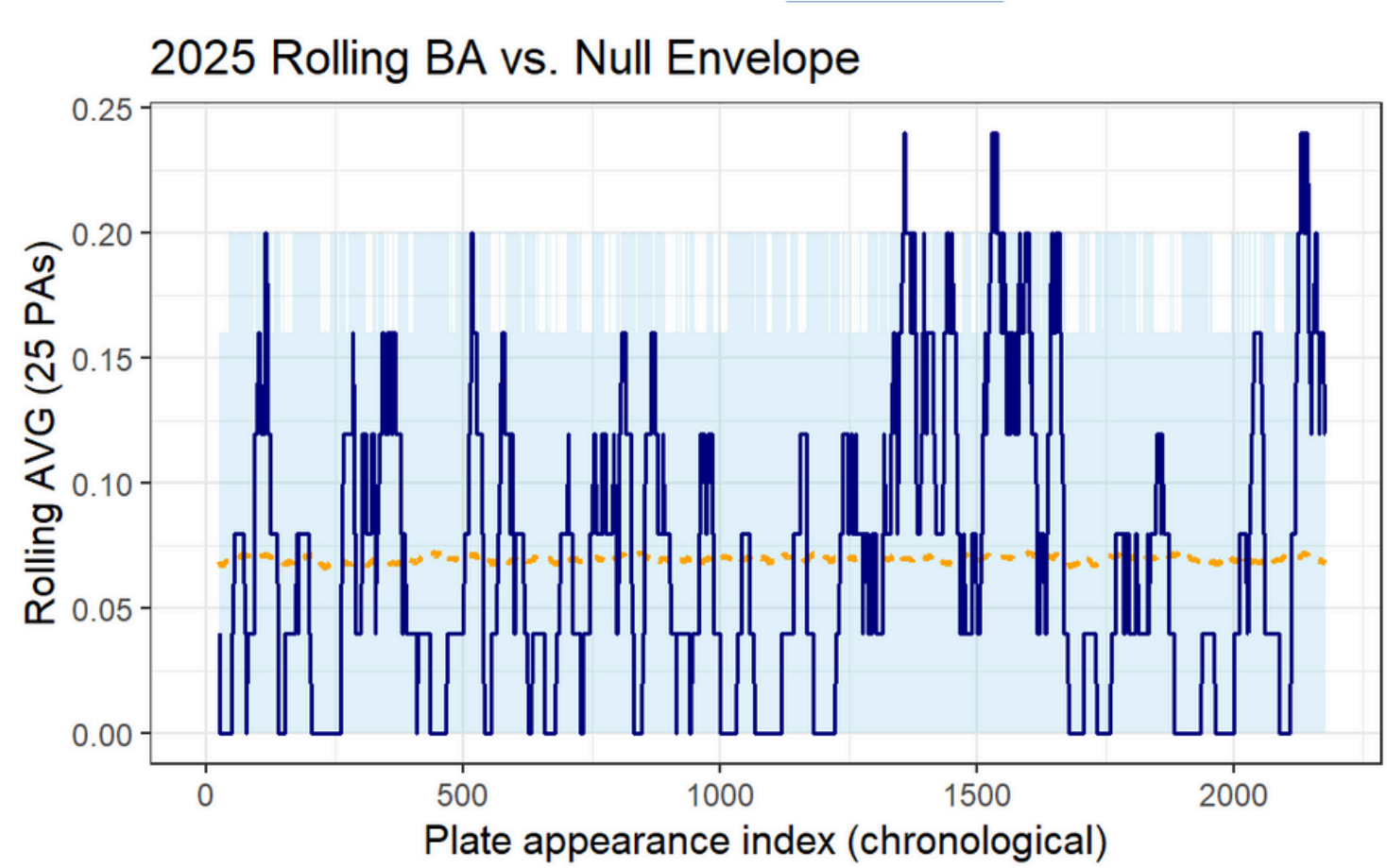
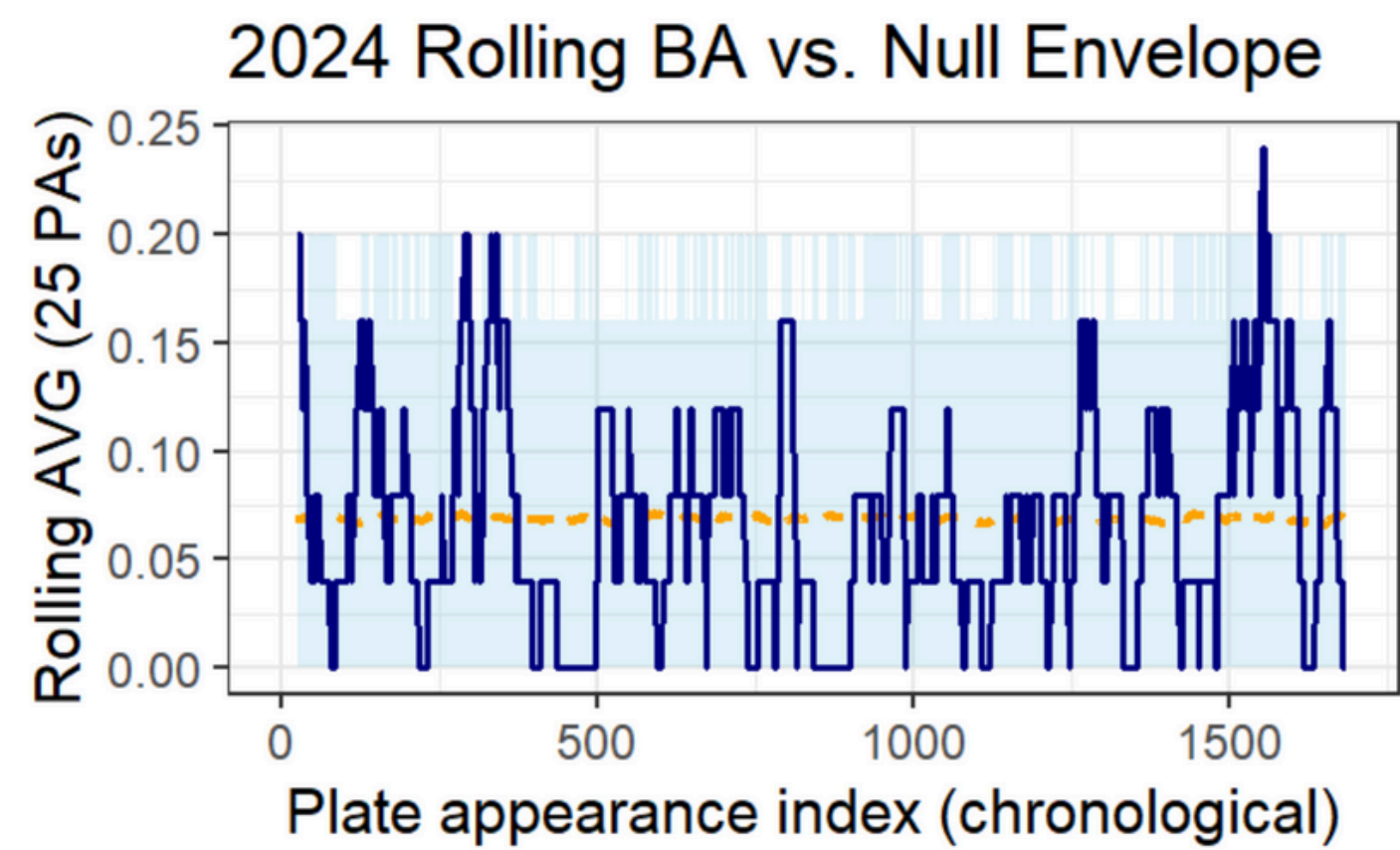
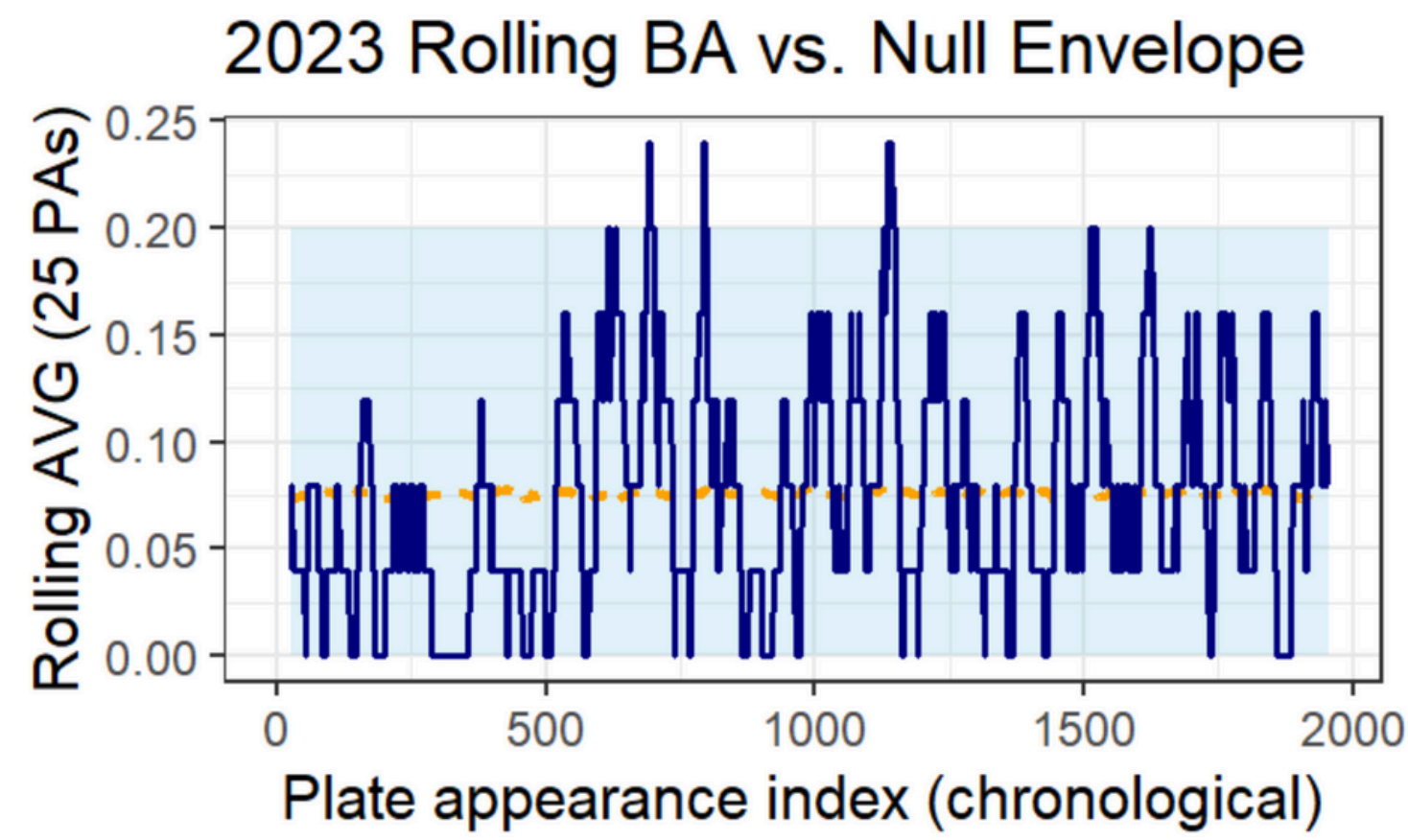
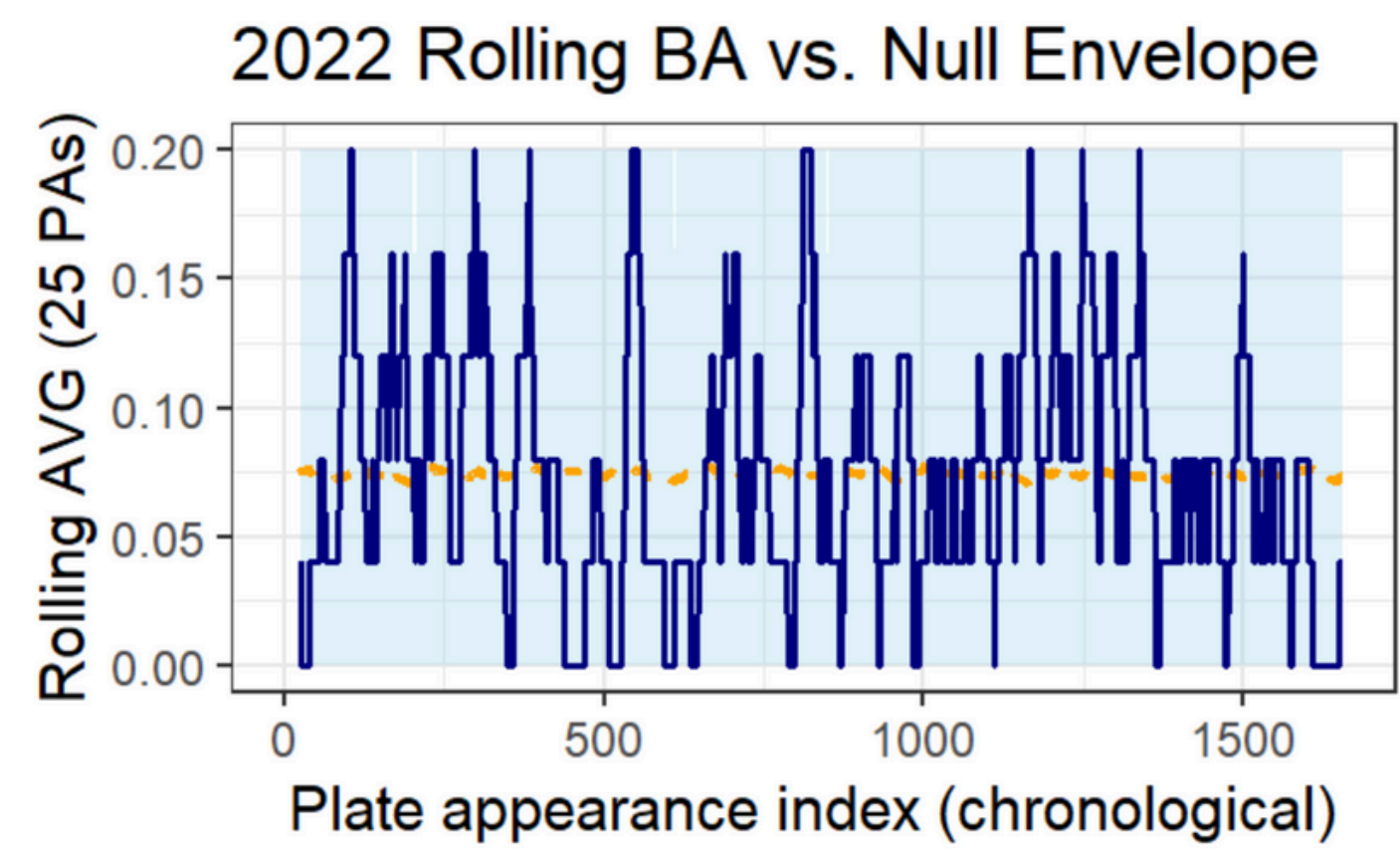
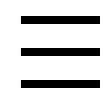
03

Batting Average by pitch location

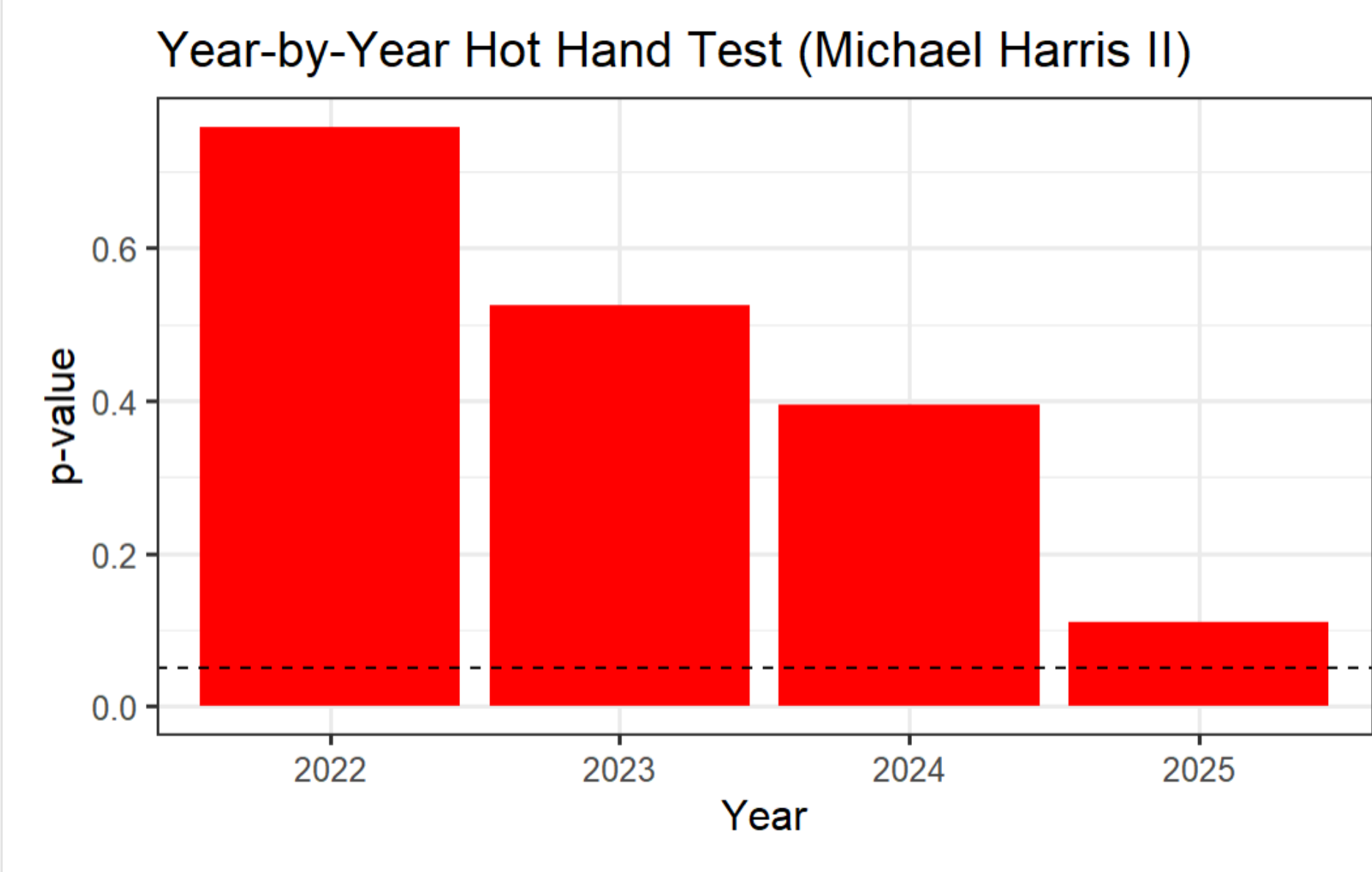
04

Simulation-based hot hand test

ROLLING BATTING AVERAGE & SIMULATION

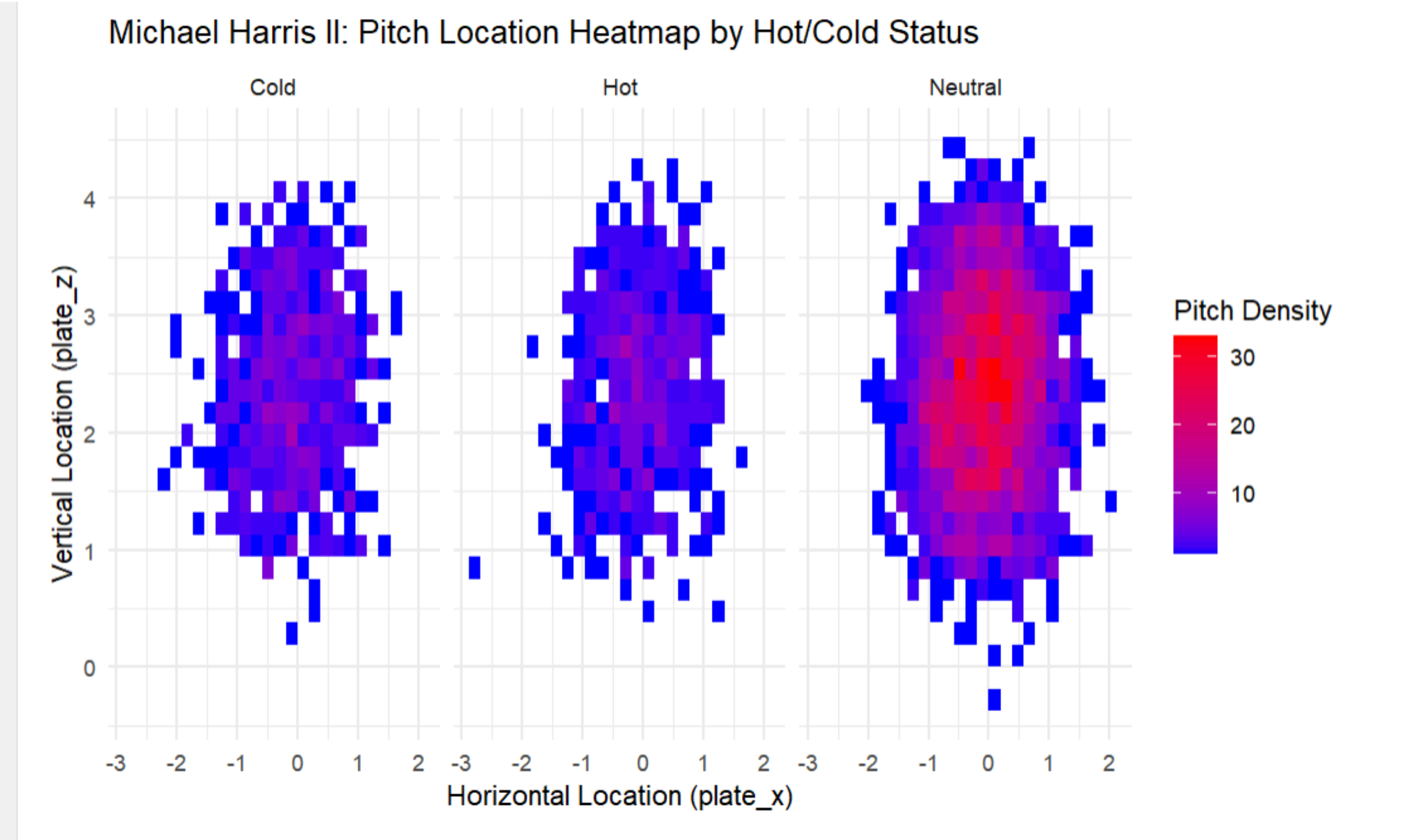


ROLLING BATTING AVERAGE & SIMULATION

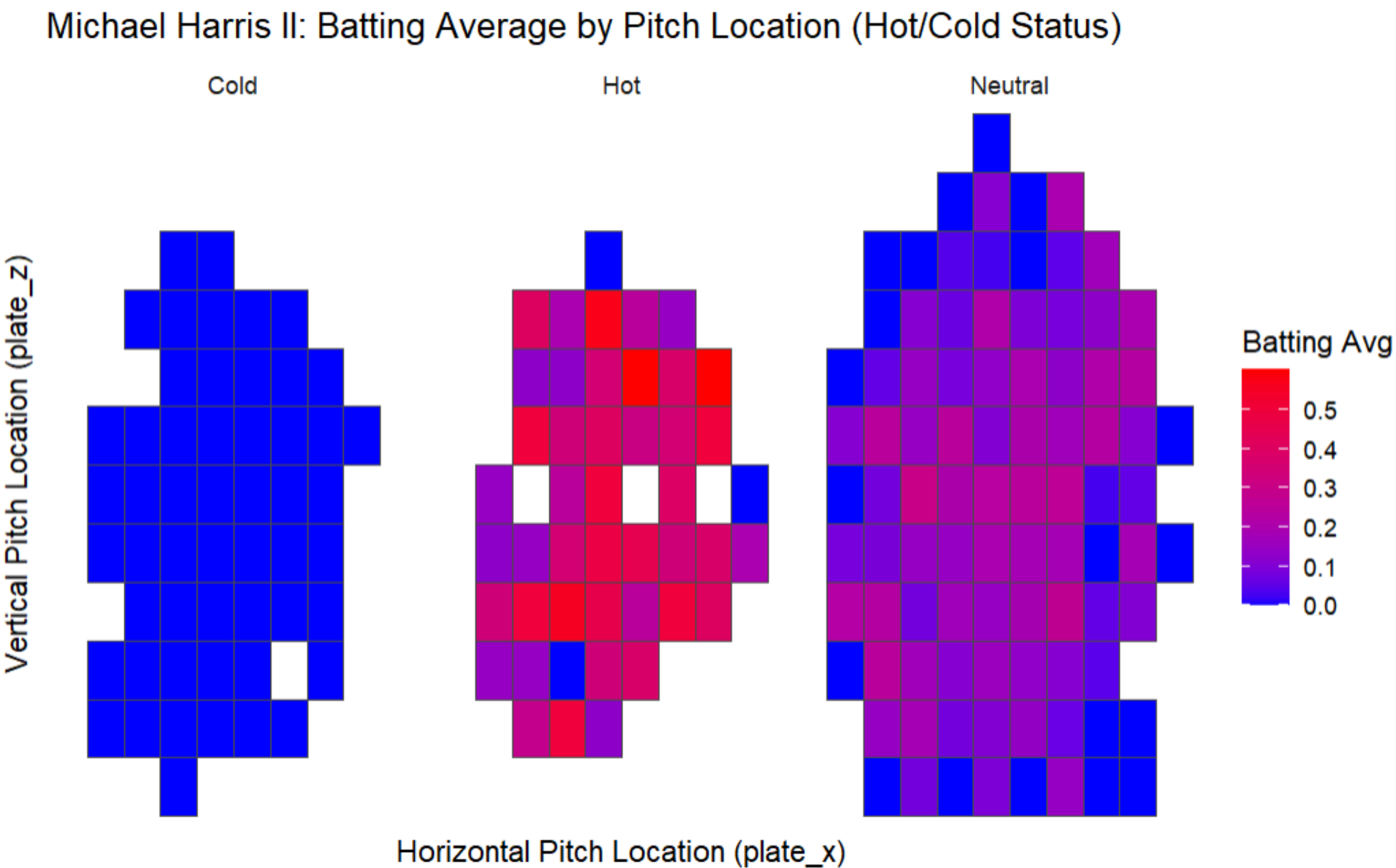


year <int>	real_var <dbl>	mean_ba <dbl>	p_value <dbl>
2022	0.004116629	0.07432024	0.7490
2023	0.004563049	0.07574207	0.5364
2024	0.004358119	0.06884273	0.3864
2025	0.004947001	0.06982085	0.1100

SWING DECISION HEATMAPS



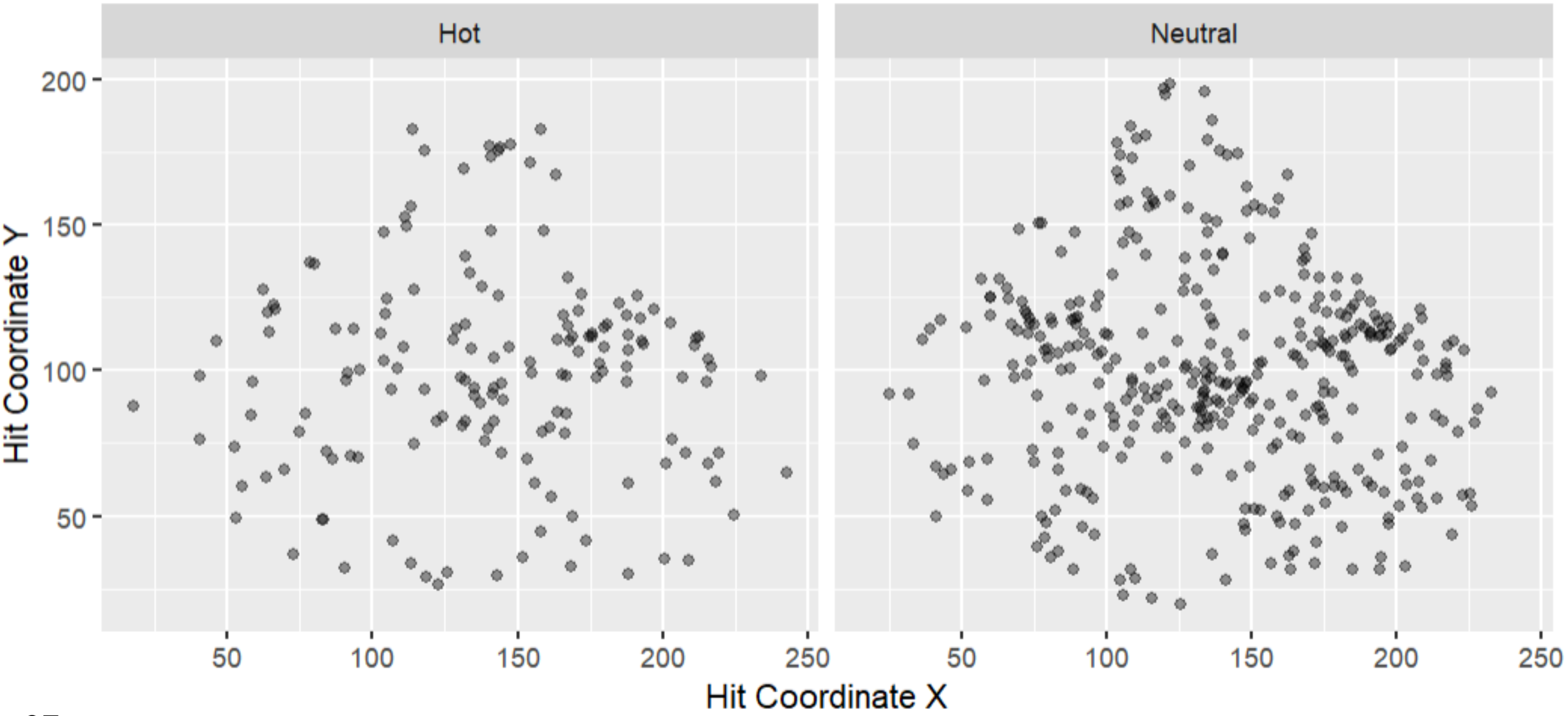
BATTING AVERAGE BY ZONE

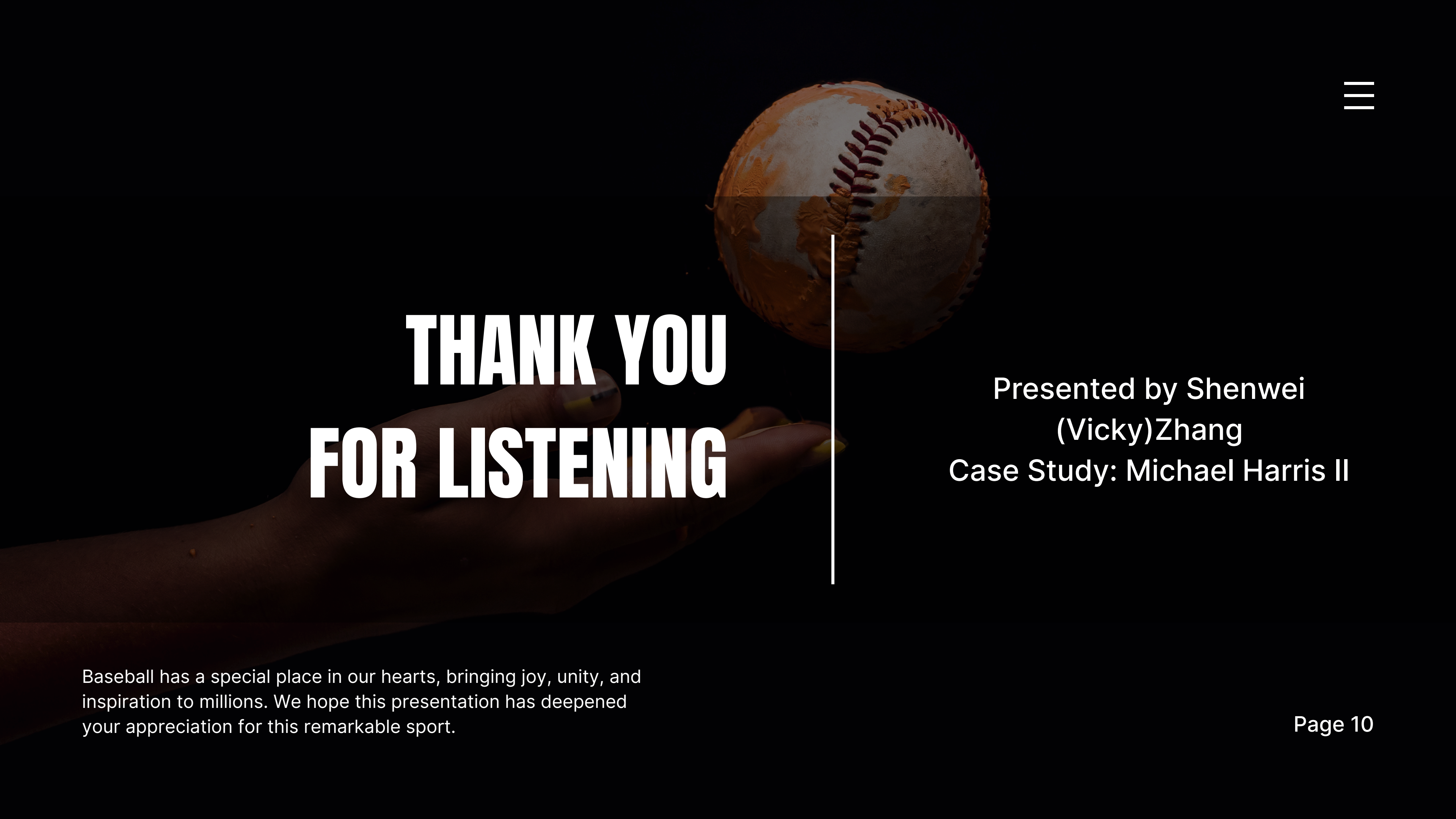


SPRAY CHART



Spray Chart by Hot/Cold Status



A hand holding a baseball with a world map painted on it. The background is dark, and the text is white.

THANK YOU FOR LISTENING

Presented by Shenwei
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Case Study: Michael Harris II

Baseball has a special place in our hearts, bringing joy, unity, and inspiration to millions. We hope this presentation has deepened your appreciation for this remarkable sport.